

BERLIN DAHLEM, Germany

WMO: 103810

Lat: 52.4536N

Lon: 13.3017E

Elev: 80

StdP: 100.37

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.0	-9.1	-15.7	1.0	-10.7	-12.8	1.3	-7.7	9.0	6.0	7.8	5.2	2.6	110	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	29.3	19.0	27.3	18.1	25.6	17.4	20.2	26.5	19.3	25.2	18.3	23.6	3.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.2	22.3	17.1	12.4	21.1	16.2	11.6	20.7	58.3	26.4	55.1	25.4	52.1	23.7	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.5	5.9	DB	-12.4	33.2	4.7	2.0	-15.8	34.6	-18.5	35.8	-21.2	36.9	-24.6	38.4	(4)
(5)			WB	-12.8	22.2	4.6	0.9	-16.1	22.8	-18.8	23.3	-21.4	23.8	-24.7	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	0.6	0.5	4.8	8.6	13.8	16.3	18.7	18.2	14.1	9.7	4.5	2.0	(6)	
	DBStd	7.56	5.42	4.77	3.68	3.73	3.88	3.54	3.30	3.31	2.74	3.64	3.43	3.87	(7)	
	HDD10.0	1270	292	268	164	69	11	1	0	0	1	50	166	249	(8)	
	HDD18.3	3390	550	501	419	291	147	81	37	45	129	268	414	507	(9)	
	CDD10.0	1039	0	1	3	28	128	190	269	253	125	41	1	1	(10)	
	CDD18.3	118	0	0	0	0	7	21	47	40	3	0	0	0	(11)	
	CDH23.3	979	0	0	0	3	59	195	406	293	23	1	0	0	(12)	
	CDH26.7	234	0	0	0	0	3	46	105	79	1	0	0	0	(13)	
(14) Wind	WSAvg	3.0	3.4	3.2	3.3	3.0	2.8	3.0	2.7	2.5	2.7	2.9	3.1	3.3	(14)	
(15) Precipitation	PrecAvg	576	46	34	39	36	53	66	63	60	45	41	47	49	(15)	
	PrecMax	835	119	90	117	88	147	193	185	160	135	144	102	130	(16)	
	PrecMin	376	6	2	9	1	4	2	7	7	5	4	1	10	(17)	
	PrecStd	108	24	21	22	21	31	42	35	33	26	29	21	23	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	14.1	17.1	22.7	27.0	30.8	31.8	32.1	26.0	21.4	13.2	11.9	(19)	
		MCWB	8.9	9.7	10.5	13.0	16.6	19.1	20.1	20.3	18.2	15.8	10.8	9.6	(20)	
	2%	DB	9.2	10.5	13.8	19.7	25.0	27.7	29.4	28.6	23.4	18.6	11.8	10.5	(21)	
		MCWB	7.4	7.9	9.0	12.1	15.5	17.9	19.0	19.3	16.8	14.0	9.9	8.7	(22)	
	5%	DB	8.1	8.2	11.7	17.3	23.2	25.6	27.5	26.6	21.0	16.4	10.5	9.3	(23)	
		MCWB	6.4	6.2	7.7	10.7	14.6	16.9	18.3	18.4	15.7	13.0	8.8	7.9	(24)	
	10%	DB	6.9	6.3	10.1	15.2	21.2	23.3	25.7	24.3	19.0	14.9	9.3	7.8	(25)	
		MCWB	5.4	4.5	7.2	9.7	13.6	15.6	17.8	17.5	14.5	11.9	7.8	6.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.3	11.4	13.8	18.3	20.6	21.8	21.5	19.0	16.7	11.5	9.9	(27)	
		MCDB	10.7	13.3	15.7	21.1	24.3	27.6	28.5	29.7	24.2	19.3	12.5	11.0	(28)	
	2%	WB	7.9	8.1	9.6	12.5	16.8	18.9	20.4	20.1	17.7	14.8	10.2	9.1	(29)	
		MCDB	9.1	10.5	12.5	18.3	23.2	25.8	26.7	26.3	21.5	17.7	11.5	10.3	(30)	
	5%	WB	6.7	6.4	8.5	11.3	15.4	17.7	19.3	19.3	16.5	13.4	9.0	7.9	(31)	
		MCDB	7.9	8.1	11.1	16.2	21.0	23.2	25.4	24.7	19.8	15.9	10.1	9.1	(32)	
	10%	WB	5.4	4.6	7.3	10.1	14.3	16.5	18.2	18.3	15.4	12.4	8.1	6.6	(33)	
		MCDB	6.7	6.1	9.9	14.6	19.9	21.8	23.9	23.0	18.5	14.4	9.1	7.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.6	6.7	8.4	9.6	9.1	9.6	9.6	8.2	7.2	4.6	3.9	(35)	
		MCDBR	4.7	7.9	9.7	11.9	12.7	13.0	13.1	12.7	11.4	9.4	5.5	4.8	(36)	
	5% WB	MCWBR	4.0	5.5	5.7	5.7	5.4	5.2	5.1	5.3	5.6	5.5	4.4	4.2	(37)	
		MCWBR	4.7	7.7	9.0	10.7	11.0	11.6	11.6	11.3	10.0	8.5	5.1	4.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.326	0.352	0.393	0.417	0.414	0.411	0.432	0.425	0.388	0.375	0.350	0.320	(40)	
		taud	2.310	2.285	2.192	2.201	2.252	2.285	2.238	2.270	2.352	2.346	2.350	2.322	(41)	
	Ebn at Noon		652	745	783	814	837	843	815	798	787	710	618	590	(42)	
		Edh at Noon	63	86	115	129	128	125	129	119	98	81	62	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.63	1.33	2.35	3.93	4.87	5.18	4.85	4.17	3.04	1.69	0.78	0.47	(44)	
	RadStd		0.07	0.23	0.26	0.51	0.58	0.41	0.64	0.34	0.35	0.20	0.10	0.05	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page